LAB EXPERIMENT RECORD

22-10-2026---Set 25: Website Traffic Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Day	THURSDAY

Aim

To analyze website traffic data in R and Tableau using a line chart, bar chart, stacked area chart and an interactive dashboard.

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

Question 1: Line Chart of Page Views (Tableau)

Question: In Tableau, create a line chart to visualize the trend in page views over time. Label the axes and title the chart.

Aim: To visualize the trend in daily page views using a Tableau line chart with labelled axes and a title.

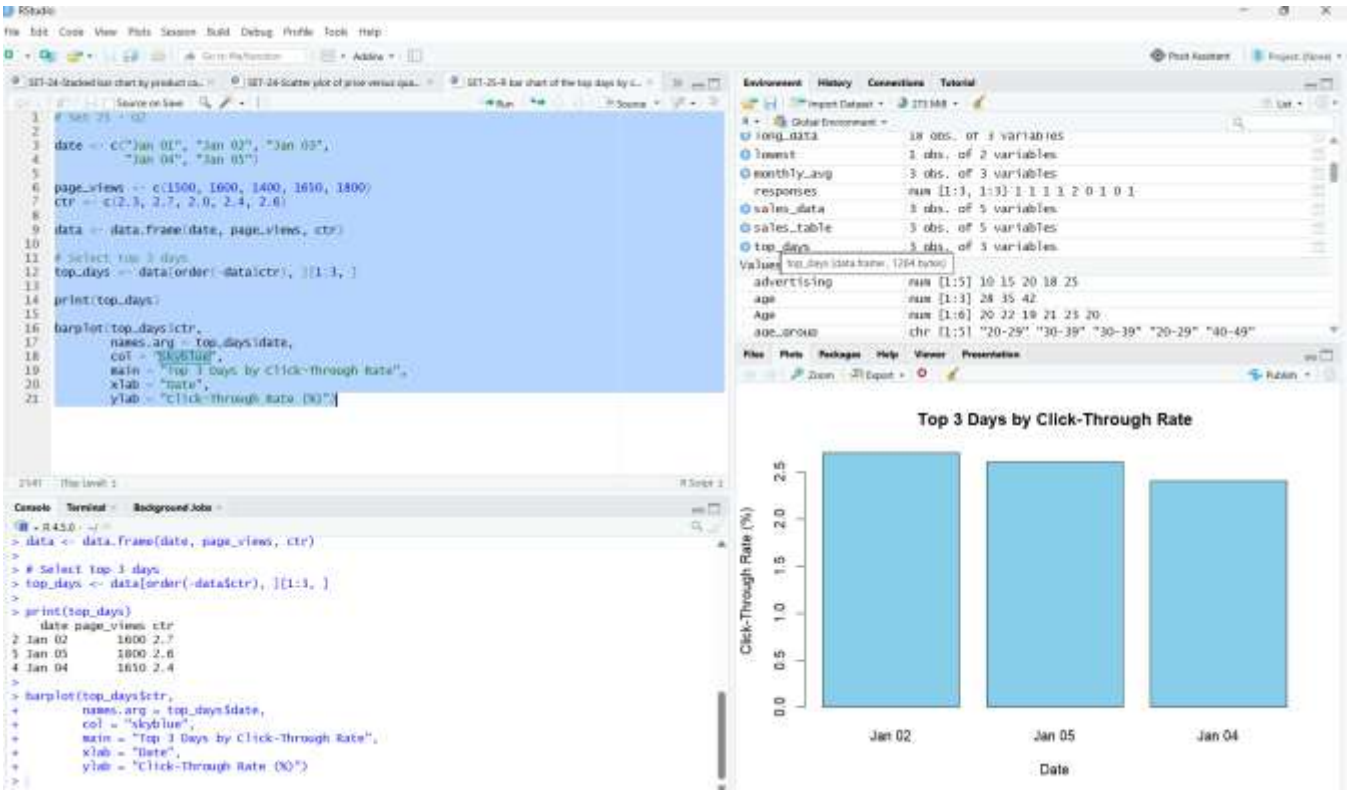


Result: Page views rise from 1500 to 1600, dip to 1400 on 2023-01-03, and then climb to 1650 and a peak of 1800 on 2023-01-05, so the overall trend is upward.

Question 2: Bar Chart of Top N Days by Click-through Rate (R)

Question: In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.

Aim: To identify the top N days with the highest click-through rates using a labelled bar chart.

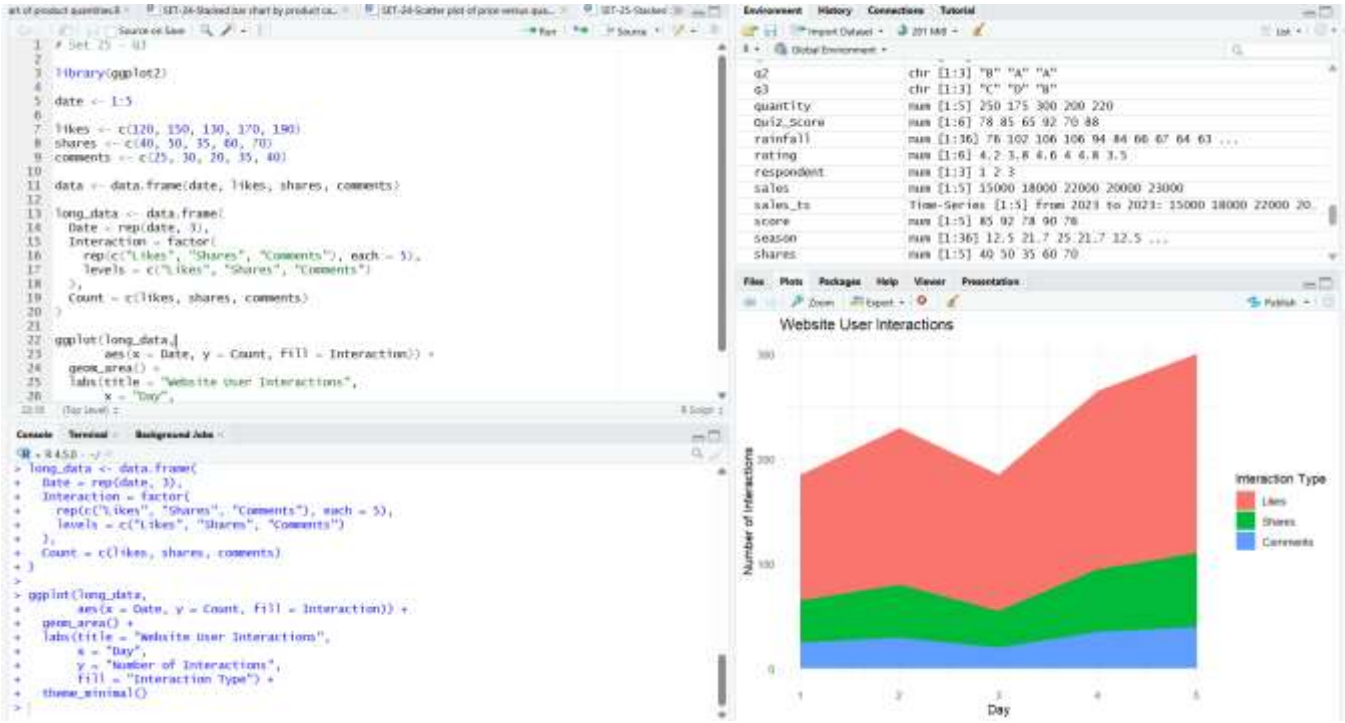


Result: For N = 3, 2023-01-02 ranks first (2.7%), followed by 2023-01-05 (2.6%) and 2023-01-04 (2.4%).

Question 3: Stacked Area Chart of User Interactions (R)

Question: In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.

Aim: To display how likes, shares and comments add up to total user interactions over time using a stacked area chart.



Result: The chart shows each interaction type as a layer stacked over the days; the dataset has no likes, shares or comments columns, so these values must be supplied.

Question 4: Tableau Dashboard with Map and Bar Chart

Question: In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart displaying page views by source.

Aim: To build an interactive Tableau dashboard that shows traffic sources on a map and page views by source in a bar chart.



Result: The map and bar chart are linked, so selecting a source on the map highlights its page views in the bar chart; the dataset has no source or location column, so these must be supplied.

Overall Result

Observation: The average page views are 1590 per day and the average click-through rate is 2.4%, with the most views on 2023-01-05 (1800) and the best click-through rate on 2023-01-02 (2.7%).